

Reading the Business Behind a Kenyan Share Price

Part 2 of 4 Financial statements, sector economics, and valuation

A company becomes easier to analyse when the investor can describe how it earns a shilling and what happens before that shilling reaches shareholders. A brewer buys inputs, produces stock, sells through a distribution network, collects payment, and funds the next production cycle. A bank gathers funding, allocates credit, earns interest and fees, absorbs losses, and maintains capital. An insurer accepts uncertain future obligations in exchange for premiums and invests funds while claims develop. These operating differences shape the financial statements and the valuation method. This article follows them through a practical research process. The accompanying workbook provides connected calculations, while the appendix explains the technical formulas and their assumptions. The objective is a valuation range whose business logic you can understand, question, and update when new evidence arrives.

Begin with a short business description before opening the ratio table. Identify the main products, customers, regions, revenue drivers, and recurring obligations. Then ask what allows the company to earn attractive returns: price, volume, distribution, scale, convenience, specialised assets, or a combination. A useful answer links the competitive advantage to an observable measure. Pricing power may appear in margins and customer retention. Efficient distribution may appear in turnover and working capital. A strong funding franchise may appear in deposit stability and funding costs. This gives the investor a way to test the business story against the statements. It also helps select comparable companies because businesses with similar labels can have very different revenue sources, capital needs, geographical exposures, and sensitivity to changes in demand or financing conditions.

The annual report can then be read as a connected account of operations and ownership. The income statement explains earnings over a period, the balance sheet records resources and obligations at a date, and the cash-flow statement follows movements in cash. Changes in equity connect profits, distributions, new capital, and other adjustments to the shareholders' position. The SEC's introductory guide offers a useful explanation of these relationships [1]. For Kenyan issuers, the investor then works with the accounting framework and policies used in the actual report. Read the auditor's opinion, key audit matters, accounting policies, and supporting notes alongside the primary statements. These sections identify where estimates, classifications, or uncertainty are important enough to affect the interpretation of an apparently simple growth rate or valuation multiple.

Build a comparable financial record

Five years of comparable information usually reveal more than a single result. Create a record of revenue, margins, profit attributable to ordinary shareholders, operating cash flow, capital expenditure, debt, equity, and shares. Preserve fiscal-year dates and distinguish consolidated group figures from the parent company's figures. Where comparatives have been restated, use a consistent series and record the reason. Acquisitions and disposals also deserve a bridge because they can change the size of the reported business without changing the performance of the original operations. The workbook includes places to record these distinctions rather than blending them into a single growth number. Once the record is consistent, the investor can ask whether improvement comes from stronger operations, a larger reporting perimeter, financing effects, tax changes, or an unusual gain.

Revenue analysis begins by separating price from volume where the company supplies enough information. Higher selling prices can support revenue while customers buy fewer units. Expansion into a new region can

create growth alongside substantial initial costs. Currency translation can change reported shillings even when local activity moves differently. A useful research note identifies the drivers that explain most of the change and tests whether they can continue. For a manufacturer, that may involve input costs, product mix, production utilisation, and distribution. For a telecommunications business, it may involve active customers, usage, service pricing, and the cost of extending the network. The modelling choice follows the evidence: use the most specific reliable drivers available, and use aggregate assumptions where a more detailed breakdown would require information that the company has not supplied.

Margins translate those revenue drivers into operating performance. Gross margin helps explain the relationship between sales and direct costs, while operating margin incorporates the broader cost of running the business. Changes in margins deserve a practical explanation. A company may improve profitability through pricing, procurement, automation, lower distribution costs, or a shift towards more profitable products. It may also benefit temporarily from lower input prices or deferred expenditure. The investor can compare management's explanation with segment disclosures and several years of results. This creates a more useful forecast than extending the latest percentage indefinitely. In the workbook, revenue growth and operating margin are independently editable by year, allowing the investor to test combinations such as slower sales with stable efficiency or stronger sales with investment-related pressure on margins.

Normalising earnings means identifying the level of profit that best represents the business being valued. Asset-disposal gains, unusual impairments, litigation, restructuring, and currency effects may deserve separate treatment, depending on their recurrence and economic significance. The adjustment should show both the reported figure and the reason for the alternative estimate. Some costs described as exceptional recur often enough to be part of the operating pattern. Some gains represent a real release of value but contribute little to next year's trading profit. Keeping these distinctions visible allows the investor to evaluate both the current result and the forward earnings base. It also makes the valuation easier to challenge: another reader can accept the reported accounts while choosing a different normalisation assumption and observing the effect on the estimated value.

Per-share analysis keeps the focus on the investor's ownership claim. Total profit can rise while earnings per share grow slowly if additional shares have been issued. A buyback can have the opposite effect, although its value depends on the purchase price and the use of cash. Dividends, earnings, book value, and valuation should therefore use consistent share definitions and periods. Basic and diluted earnings may differ where potential shares exist, and a closing share count can differ from the weighted average used for reported earnings. The workbook records units explicitly so millions of shillings and millions of shares produce shillings per share. This is a small technical detail with a large practical effect: an incorrect share denominator can make an otherwise thoughtful valuation appear dramatically cheaper or more expensive than it really is.

Follow profit into cash and capital

Operating cash flow explains how the accounting result connects to cash generated by activity. IAS 7 classifies cash flows into operating, investing, and financing categories, with the relevant accounting policies affecting the presentation [2]. For a nonfinancial company, changes in receivables, inventory, and payables are often central to the bridge. A profitable expansion may initially consume cash as stock and customer credit grow. A decline in activity may temporarily release cash by reducing those balances. The investor can therefore compare cash conversion across several years and examine the underlying operating movements. This turns a cash-flow ratio into a business explanation. It also provides useful forecast drivers because future growth will often require additional working capital before the new sales have been fully collected.

Equation (2.1), operating working capital = trade receivables plus inventory minus trade payables, creates a manageable starting point for a manufacturer. The next step is to connect each balance to activity. Receivable days estimate how long sales remain unpaid, inventory days estimate how long stock is held, and payable days estimate how long suppliers finance purchases. When purchases are unavailable, cost of sales can serve as an explicitly labelled proxy for payable days. Average balances improve the comparison where opening and closing values are available. Seasonal businesses may require more frequent observations because a year-end balance can be unrepresentative. The appendix explains the exact formulas and manual calculations. In the body of the analysis, the useful question is how much additional cash the company needs when sales or payment behaviour change.

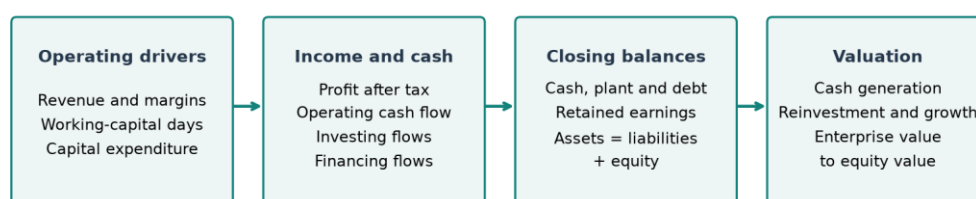
A negative cash-conversion cycle can be commercially attractive because customers pay before the company settles the associated supplier obligations. Understanding its cause is essential to assessing durability. Prepaid sales and reliable supplier terms can support a stable pattern, while overdue payments or temporary inventory reductions can produce a similar numerical result through a different process. Supplier-finance arrangements also deserve separate attention because they can change the interpretation of payables and cash flows. The sector-screen sheet lets you compare an efficient cycle, slower collections, and longer supplier credit using the same formulas. You can then ask which changes reflect a repeatable operating advantage and which represent financing pressure. The conclusion feeds directly into the cash forecast, dividend capacity, and the amount of financial flexibility available during a difficult trading period.

Capital expenditure links today's cash to tomorrow's productive capacity. Maintenance spending preserves existing operations, while expansion spending aims to create additional earnings. Companies may provide useful information about projects, capacity, network investment, or replacement programmes, although the split between maintenance and growth is often an analytical estimate. Record that estimate and test alternatives. Depreciation offers context about past capital consumption, but future replacement costs and technology can change the cash required. Equation (2.2), cash after capital expenditure = operating cash flow minus capital expenditure, provides a simple cash measure for the initial review. The valuation model then develops an unlevered cash-flow measure with consistent tax and financing treatment. The distinction helps the investor avoid mixing cash available after interest with a discount rate designed for cash available to all capital providers.

Debt analysis turns the balance-sheet total into a schedule of obligations. Record the currency, interest structure, maturity, security, covenants, and access to refinancing for material facilities. A company with moderate debt and a concentrated near-term maturity can face a different problem from one with higher debt spread across several years. Foreign-currency borrowing adds another sensitivity when operating cash is mainly in shillings. Cash balances also need classification because some may be restricted or held in subsidiaries with distribution constraints. The integrated teaching model uses a simple debt schedule with interest on opening debt and explicit principal repayments. This keeps the cash consequences visible. A company-specific version can extend that schedule to multiple facilities, changing rates, and refinancing assumptions as the relevant disclosures become available and material to the valuation.

The balance sheet provides the reconciliation that ties these decisions together. In the teaching model, plant rolls forward through capital expenditure and depreciation, debt rolls through repayments, equity rolls through profit and dividends, and cash rolls through operating, investing, and financing flows. The resulting assets equal liabilities plus equity in every forecast year. Cash emerges from the operating and financing assumptions rather than being inserted to force that equality. This is useful because an ambitious dividend or investment plan can now reveal a funding shortfall. The investor can respond by changing the planned distribution, financing, growth, or spending assumptions and examining the consequences. A connected model turns an appealing business story into a set of commitments that must work together over time, making the valuation more informative and easier to audit.

Connect operating assumptions to a complete financial model



Fictional manufacturer, KES million. The Statements sheet checks the balance sheet in each forecast year.

statements flow

Analyse banks through losses, funding, and capital

For a bank, asset quality is a central link between current earnings and future distributions. Read the gross loan book, nonperforming loans, impairment allowances, write-offs, recoveries, and movement between credit-quality categories. IFRS 9 provides the expected-credit-loss framework used in the accounting analysis [3]. The practical question is how changes in borrower performance and economic expectations affect the profit available to shareholders. A falling nonperforming-loan ratio can reflect collections, write-offs, loan growth, or changes in classification. Each route has different implications. The workbook's banking analysis therefore uses several measures together and records their definitions. This helps the investor connect the stock of troubled assets with the current impairment charge and the remaining exposure, producing a clearer view of the bank's capacity to absorb further deterioration while supporting its business.

Coverage ratios are most useful when the numerator and denominator are precisely matched. An investor might compare impairment allowances with nonperforming loans, but published definitions can include or exclude collateral, interest in suspense, regulatory reserves, and different categories of exposure. Record the chosen basis and use it consistently across periods and peers. The current credit-loss charge is a flow, while the allowance is a balance-sheet stock, so the two answer related but separate questions. A practical review follows the allowance from opening balance through new charges, recoveries, write-offs, currency effects, and closing balance where disclosed. This can reveal why a lower charge coexists with a significant remaining problem book. It also makes stress testing more useful because an additional loss assumption can be connected to earnings and capital rather than applied as an isolated percentage.

Funding and efficiency determine how much income is available before credit losses. Net interest income reflects asset yields, funding costs, and the balance-sheet mix. Fee income can diversify revenue but may have its own volume and regulatory sensitivities. The cost-to-income ratio then shows how operating expenses relate to the selected income measure. A useful bank forecast separates revenue growth, operating costs, and impairment rather than collapsing them into one profit-growth assumption. It can test lower lending yields alongside sticky deposit costs, slower fees, or a rise in collection expenditure. The investor should also examine deposit concentration and maturity information because funding stability affects the bank's room to respond. These operating relationships explain how a bank can remain profitable while its distributable capacity changes materially between otherwise similar reporting periods.

Capital connects the bank's balance sheet with its ability to grow and distribute. The relevant ratio compares eligible regulatory capital with risk-weighted assets, using the rules applicable to the regulated entity. CBK's supervision reports provide regulatory context, while the issuer's disclosures establish the actual entity-level position [4]. A regional group may have subsidiaries governed by different requirements, so group equity and a single headline ratio need an appropriate bridge. The teaching model makes its

simplifying assumption explicit: opening equity is used as a proxy for eligible capital. It then calculates the retention needed to support assumed growth in risk-weighted assets. The result helps explain why a strong accounting profit can be divided differently between dividends and retained capital as lending opportunities, loss expectations, and required buffers change.

Read insurers and asset-heavy businesses on their own terms

Insurance combines underwriting obligations with investment activity, and the two need a clear bridge to shareholder profit. For a short-tail non-life illustration, the loss ratio and expense ratio can be added when both use the same earned-premium denominator. The resulting combined ratio explains the underwriting result before investment and other items. Published IFRS 17 reporting requires attention to insurance revenue, service expenses, reinsurance, and the issuer's own performance definitions [5]. Life insurance introduces further features, including the release of profit as services are provided. The workbook therefore presents its earned-premium example as a defined teaching model and provides a review field for mapping a real issuer's disclosures. This allows the investor to use the arithmetic confidently while preserving the accounting and business distinctions that determine what a published ratio actually measures.

Equation (2.3), shareholder profit = underwriting result plus investment and other results minus financing costs and tax, organises the simplified insurance bridge. Return on equity then compares attributable profit with average equity over the period. A strong investment year can support profit even when underwriting is weak, while market losses can reduce returns during an otherwise sound underwriting period. Net asset value requires its own examination of investment valuations, liabilities, minority interests, and the ownership structure. Jubilee's annual report provides separate group and company information and notes explaining earnings per share and dividends [6]. The practical research task is to identify which earnings and assets belong to the listed shareholders and how cash can move to them. That ownership perspective makes a price-to-book comparison more informative than treating every shilling of consolidated assets as equally distributable.

Agriculture, energy, and investment holding companies extend the same principle of matching the method to the business. Agricultural analysis follows crop volumes, realised prices, weather, seasonal working capital, and the distinction between cash earnings and biological-asset valuation movements. Energy analysis follows capacity, output, tariffs, offtaker payments, maintenance, project commitments, and financing currency. Holding-company analysis looks through to the underlying assets, then subtracts central debt, costs, taxes, and other leakage where relevant. These subjects belong in the research log even when the initial portfolio contains only a small allocation to the sector. They identify the assumptions most likely to change value. The investor can then build a focused model around those drivers instead of applying a bank or manufacturer template mechanically to a business with a different operating and ownership structure.

Estimate value from cash generation and required returns

Valuation begins by asking what the investor expects to receive and what return compensates for committing money to that opportunity. For the fictional manufacturer, the workbook forecasts operating cash available to all capital providers after tax and reinvestment, then discounts those cash flows at a matching required return. A terminal calculation represents the business beyond the explicit forecast. Cash and nonoperating assets are added where appropriate, debt and other claims are deducted, and the result is divided by shares. The appendix contains the detailed discounted-cash-flow formulas and every step of the worked example. The article's practical focus is the set of business assumptions behind them: sales, margins, tax, working capital, productive investment, financing, and the returns the company can earn as it moves towards a more mature operating pattern.

The terminal period deserves particular care because growth requires resources. Damodaran's valuation material connects stable growth to the reinvestment needed at the assumed return on capital [7]. In the workbook, a 4% perpetual growth assumption and a 16% return on new capital imply retaining 25% of after-tax operating profit for reinvestment. The remaining portion supports the terminal cash-flow calculation. Raising growth therefore changes both the future profit level and the resources required to sustain it. This is economically more informative than increasing growth while leaving investment unchanged. The sensitivity grid recalculates the full relationship across required returns and growth rates. It gives the investor a range to interpret and makes the assumptions behind a high terminal value visible, including how much of the total valuation depends on events beyond the explicit forecast.

Banks and insurers often benefit from valuation methods centred on equity, distributions, and returns on equity, as developed in Damodaran's treatment of financial-service firms [8]. The workbook includes a steady-state price-to-book model that connects sustainable return on equity, growth, and the shareholder's required return. It also includes a separate near-term capital-capacity calculation. Together, these provide two useful views: what a mature stream of equity earnings might be worth and how much current profit can be distributed while supporting the assumed capital requirement. A price below book value can be attractive when future returns justify more than the market price. It can also reflect low earning power or risks within the assets. The investor's task is to explain which interpretation fits the evidence and how sensitive the value is to a different outcome.

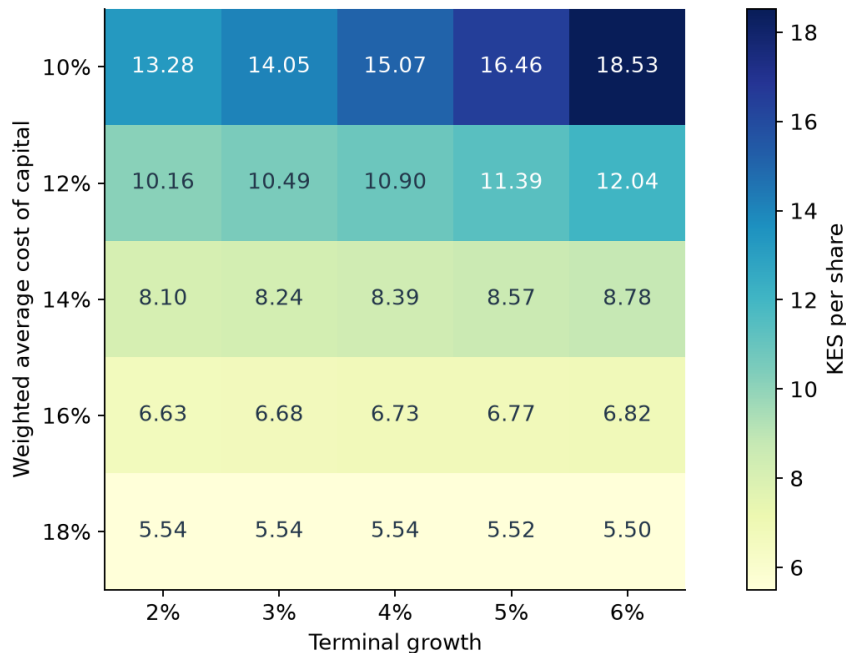
Market multiples offer a useful cross-check when their inputs and peer groups are consistent. Price-to-earnings compares the equity price with earnings available to that ownership claim. Enterprise-value multiples compare operating value with operating measures before financing. Price-to-book is often relevant where assets and capital play a central role in generating earnings. Each multiple carries assumptions about growth, profitability, reinvestment, risk, and accounting quality. The investor can compare a company with peers and its own history, then explain the reason for any premium or discount. This keeps the multiple connected to the business. A low multiple becomes a research question about the durability of earnings and the required return, while a high multiple becomes a question about the growth and quality already embedded in the purchase price.

Equation (2.4), margin of safety = one minus purchase price divided by estimated value, expresses the difference between price and a selected valuation estimate. The estimate should come from a range of coherent cases. A base case might assume steady demand and normal collections, while a weaker case combines slower sales, narrower margins, and delayed customer payments. A stronger case should also fund the investment needed to support its growth. The investor can compare the proposed entry price across all three cases and decide how much uncertainty the position size can accommodate. This turns a margin-of-safety percentage into a practical judgement about resilience. The accompanying sensitivity tables then identify which inputs deserve the most research attention because small changes in them produce large differences in estimated value.

Financial scoring can help organise that attention. Piotroski's peer-reviewed research examined how historical statement signals helped distinguish outcomes within a high book-to-market investment setting [9]. Its broader lesson for this exercise is the usefulness of combining several aspects of financial change rather than relying on one attractive ratio. The workbook's review process therefore looks at profitability, cash conversion, financing, efficiency, and ownership together, with sector-specific interpretation. A score can summarise the observations, but the supporting evidence remains valuable when a company is unusual or undergoing change. The investor can document why a weaker recent metric might improve, what would confirm that improvement, and how much capital to commit before it becomes visible. This creates a disciplined research process that remains open to business developments instead of treating the screen as the final judgement.

The final company note should connect the research to a decision. Describe how the business earns money, the financial evidence supporting that view, the valuation range, and the assumptions that would materially change it. Add the expected source of shareholder return, the major risks, and the conditions that would lead to further buying, a smaller position, or an exit. The workbook provides the calculations, while the note preserves the reasoning that makes those calculations useful. You now have more than a dividend candidate: you have an ownership case with a price range and a clear account of uncertainty. The next article combines several such cases into portfolios, where the interaction between holdings, household needs, and market liquidity becomes as important as the attractiveness of each company considered separately.

Estimated value per share changes with the assumptions



Fictional manufacturer; terminal return on invested capital 16%. Central case: KES 8.39 per share.

dcf sensitivity

References

[1] U.S. Securities and Exchange Commission, “Beginners’ Guide to Financial Statements”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[2] IFRS Foundation, “IAS 7 Statement of Cash Flows”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[3] IFRS Foundation, “IFRS 9 Financial Instruments”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[4] Central Bank of Kenya, “Bank Supervision Annual Report 2024”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[5] IFRS Foundation, “IFRS 17 Insurance Contracts”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[6] Jubilee Holdings, “2025 Annual Integrated Report”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[7] A. Damodaran, “Excess Returns and Terminal Value”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[8] A. Damodaran, “Valuing Financial Service Firms”. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

[9] J. D. Piotroski, “Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers”, Journal of Accounting Research, vol. 38, suppl., pp. 1–41, 2000. [Online]. Available: [Source](#). Accessed: Sep. 7, 2026.

Appendix 2 Statement models, financial quality, and valuation

The manufacturer, bank, and insurance scenarios are fictional teaching cases. Their figures are not valuations of the named NSE issuers in the income screen. All currency amounts below are KES million unless identified as per-share values. Excel formulas and mathematical equivalents appear in the generated dictionary at the end.

A2.1 The integrated manufacturer

Opening balances at 31 December 2026 are revenue 10,000; receivables 1,200; inventory 1,000; payables 900; net plant 6,000; cash 1,200; debt 2,500; and equity 6,000. Assets = $1,200 + 1,000 + 6,000 + 1,200 = 9,400$. Liabilities plus equity = $900 + 2,500 + 6,000 = 9,400$. Shares outstanding are 1,000 million.

The forecast assumptions have a separate input for every year 2027–2031. Annual growth, EBITDA margin, depreciation, capex, tax, collection days, inventory days, supplier days, cost of sales, interest, principal repayment, and payout all feed the statements. Interest uses opening debt, payments occur at year-end, and the case has no new borrowing, share issuance, acquisitions, leases, tax-loss carryforwards, or other comprehensive income. Those features can be added as separate schedules when modelling an actual company.

For 2027:

$$\begin{aligned}\text{Revenue} &= 10,000 \times 1.06 = \mathbf{10,600} \\ \text{COGS} &= 10,600 \times 0.60 = \mathbf{6,360} \\ \text{EBITDA} &= 10,600 \times 0.22 = \mathbf{2,332} \\ \text{Depreciation} &= 6,000 \times 0.10 = \mathbf{600} \\ \text{EBIT} &= 2,332 - 600 = \mathbf{1,732} \\ \text{Interest} &= 2,500 \times 0.12 = \mathbf{300} \\ \text{Profit before tax} &= 1,732 - 300 = \mathbf{1,432} \\ \text{Tax} &= 1,432 \times 0.30 = \mathbf{429.60} \\ \text{Net profit} &= 1,432 - 429.60 = \mathbf{1,002.40}\end{aligned}$$

EBITDA is earnings before interest, tax, depreciation, and amortisation; EBIT is operating profit after depreciation in this case. The model forecasts EBITDA from a margin; COGS is used to determine the working-capital balances. Remaining operating costs are implicit in the EBITDA margin rather than separately forecast.

A2.2 Working capital and cash conversion

Article equation (2.1):

$$\begin{aligned}\text{NWC} &= \text{trade receivables} + \text{inventory} - \text{trade payables} \\ \text{AR} &= \text{revenue} \times \text{collection days} / 365 \\ \text{Inventory} &= \text{COGS} \times \text{inventory days} / 365 \\ \text{AP} &= \text{COGS} \times \text{supplier days} / 365 \\ \Delta \text{NWC} &= \text{closing NWC} - \text{opening NWC}\end{aligned}$$

The forecast uses closing operating balances, whereas the historical sector screen uses averages to calculate observed days. COGS is explicitly used as a purchases proxy for supplier days. Exclude cash, interest-bearing debt, and nontrade balances from operating NWC unless their economic role justifies a separately documented adjustment.

$AR_{2027} = 10,600 \times 44/365 = \mathbf{1,277.80822}$
 $Inventory_{2027} = 6,360 \times 60/365 = \mathbf{1,045.47945}$
 $AP_{2027} = 6,360 \times 50/365 = \mathbf{871.23288}$
 $NWC_{2027} = 1,277.80822 + 1,045.47945 - 871.23288 = \mathbf{1,452.05479}$
 $NWC_{2026} = 1,200 + 1,000 - 900 = \mathbf{1,300}$
 $\Delta NWC = \mathbf{152.05479}$

Historical cash-conversion-cycle formulas are $DIO = \text{average inventory}/COGS \times 365$; $DSO = \text{average trade receivables}/\text{revenue} \times 365$; $DPO = \text{average trade payables}/COGS \times 365$; and $CCC = DIO + DSO - DPO$. For the efficient-cycle example, $DIO = 500/6,000 \times 365 = 30.4167\text{days}$; $DSO = 300/10,000 \times 365 = 10.95\text{days}$; $DPO = 1,200/6,000 \times 365 = 73\text{days}$. Thus $CCC = -31.6333\text{days}$. Investigate whether the supplier terms, collections, and stock levels can persist.

A2.3 Cash, plant, debt, and equity roll-forwards

$CFO = \text{net profit} + \text{depreciation} - \Delta NWC$
 $CFO_{2027} = 1,002.40 + 600 - 152.05479 = \mathbf{1,450.34521}$
 $Capex_{2027} = 10,600 \times 0.09 = \mathbf{954}$
 $\text{Cash after capex} = 1,450.34521 - 954 = \mathbf{496.34521}$
 $\text{Dividend} = 1,002.40 \times 0.40 = \mathbf{400.96}$
 $\text{Closing cash} = 1,200 + 1,450.34521 - 954 - 250 - 400.96 = \mathbf{1,045.38521}$
 $\text{Closing plant} = 6,000 + 954 - 600 = \mathbf{6,354}$
 $\text{Closing debt} = 2,500 - 250 = \mathbf{2,250}$
 $\text{Closing equity} = 6,000 + 1,002.40 - 400.96 = \mathbf{6,601.44}$

Article equation (2.2) is the cash-after-capex calculation. In this example CFO includes interest through net profit. It is distinct from unlevered free cash flow used in enterprise valuation.

The 2027 balance sheet reconciles: assets = $1,277.80822 + 1,045.47945 + 6,354 + 1,045.38521 = \mathbf{9,722.67288}$. Liabilities plus equity = $871.23288 + 2,250 + 6,601.44 = \mathbf{9,722.67288}$. The model repeats the roll-forward through 2031, with a residual tolerance for floating-point arithmetic. A negative calculated cash balance represents an unfunded plan requiring an explicit financing or operating response.

A2.4 Unlevered cash flow and DCF

Symbol	Meaning
FCFF	Free cash flow to the firm, before financing payments
NOPAT	Operating profit after the assumed operating tax
w	Weighted average cost of capital, nominal KES
g	Stable terminal growth, nominal KES
ROIC	Return on newly invested operating capital in the terminal state
EV	Enterprise value
NCI	Minority interests or other ownership claims deducted in the equity bridge

$NOPAT = EBIT(1 - \text{tax rate})$
 $FCFF = NOPAT + \text{depreciation} - \text{capex} - \Delta NWC$
 $FCFF_{2027} = 1,732 \times 0.70 + 600 - 954 - 152.05479 = \mathbf{706.34521}$

$PV_forecast = \sum_{t=1...5} FCFF_t / (1+w)^t$
 Terminal reinvestment fraction = $g/ROIC$
 $FCFF_6 = NOPAT_5(1+g)(1-g/ROIC)$
 $TV_5 = FCFF_6 / (w-g)$
 $EV = PV_forecast + TV_5 / (1+w)^5$
 Equity value = EV + excess cash + nonoperating investments – debt – NCI
 Value per share = equity value / shares

At $w=14\%$, $g=4\%$, $ROIC=16\%$, terminal reinvestment is 25% . The workbook calculates PV_forecast **3,358.04817**, terminal NOPAT **1,626.33215**, terminal FCFF **1,219.74911**, terminal EV **12,197.49109**, and terminal PV **6,334.99466**. EV is **9,693.04283**. Adding 1,200 cash and deducting 2,500 debt gives equity value **8,393.04283**, or **KES 8.39304/share**.

Article equation (2.4): $MOS = 1 - \text{price}/\text{value}$. At illustrative price 6, $MOS = 1 - 6/8.39304 = \mathbf{28.5122\%}$. Terminal PV represents **65.3561%** of EV. The sensitivity grid recomputes both forecast discounting and terminal reinvestment for each pair of w and g . Require $w > g$, $ROIC > 0$, positive shares, and economically supportable growth; a terminal retention fraction above 100% needs a different steady-state structure.

A2.5 Bank diagnostics and distribution capacity

NPL ratio = nonperforming loans / gross loans
 Coverage = impairment allowance / NPL stock
 Cost of risk = annual impairment charge / average gross loans
 NIM = net interest income / average earning assets
 CIR = operating expense / operating income
 Capital ratio = eligible regulatory capital / risk-weighted assets
 Headroom = actual capital ratio – applicable requirement

For the fictional diagnostic base: NPL ratio = $8,000/80,000 = 10\%$; coverage = $5,000/8,000 = 62.5\%$; cost of risk = $1,200/77,000 = 1.5584\%$; NIM = $6,000/95,000 = 6.3158\%$; CIR = $3,500/10,000 = 35\%$. Core capital ratio = $11,000/80,000 = 13.75\%$, giving 3.25 percentage points over a 10.5% input. Total capital ratio = $15,000/80,000 = 18.75\%$, giving 4.25 points over 14.5%. Requirements are entity-specific inputs; the exercise is not a regulatory determination for a named bank.

For the capacity model:

Required retained capital = $\max[0, RWA_0(1 + \text{growth_RWA}) \times \text{required ratio} - \text{eligible capital}_0]$
 Dividend capacity = $\max[0, \text{scenario profit} - \text{required retained capital}]$

The proxy assumes opening equity equals eligible capital and omits regulatory deductions, additional buffers, supervisory restrictions, and subsidiary-transfer constraints. Add those explicitly for an actual bank.

The separate steady-state valuation uses $P/B = (ROE - g) / (k_e - g)$, derived from dividend retention g/ROE under clean-surplus assumptions. At $ROE 18\%$, $g 5\%$, $k_e 16\%$, $P/B = 13/11 = \mathbf{1.1818 \text{ times}}$. Opening equity 10,000 gives value 11,818.18. The long-run ROE assumption is separate from the one-year profit stress, which may differ during the transition. A multi-stage residual-income model values $B_0 + \sum [(ROE_t - k_e) B_{t-1} / (1 + k_e)^t]$ plus any terminal residual-income value.

A2.6 Insurance bridge and asset value

Article equation (2.3), for the defined non-life example:

Combined ratio = claims/earned premium + expenses/earned premium
 Underwriting result = earned premium – claims – expenses

$PBT = \text{underwriting} + \text{investment} + \text{other} - \text{finance cost}$

$PAT = PBT - \max(PBT, 0) \times \text{tax rate}$

$ROE = PAT / [(\text{opening equity} + \text{closing equity}) / 2]$

At premiums 4,000, claims 2,600 and expenses 1,080, the combined ratio is 92% and underwriting profit 320. Add 880 investment result and deduct 100 finance cost: PBT 1,100. Tax 330 leaves PAT 770; average equity 10,250 gives **7.5122% ROE**. This is a complete simplified profit bridge; investment income alone does not determine underwriting quality. Map IFRS 17 service results and reinsurance to the actual issuer's definitions before applying it.

Adjusted NAV can be expressed as reported attributable equity plus documented asset-value adjustments minus additional liability or tax adjustments. Price/adjusted book then compares market equity value with that estimate. Avoid adding the value of a subsidiary twice when it is already consolidated. For life insurers, analyse the contractual service margin, reserve assumptions, cash emergence, and capital requirements separately from a short-tail combined ratio.

A2.7 Historical quality signals

The workbook's five-year history is fictional and separate from the forecast. Its nine-signal screen is inspired by Piotroski but uses closing-asset denominators and the stated simplified definitions. It tests positive profitability, positive cash generation, improving profitability, cash exceeding accrual profit, lower leverage, improved liquidity, no share issuance, improved gross margin, and improved asset turnover. Each condition contributes 0 or 1. This educational variant should not be presented as an exact replication of the original empirical strategy.

Workbook formula dictionary

Dictionary entries

Each entry defines a formula family. Copied years, issuers, and simulation paths follow the same relationship. The cell-by-cell formula audit provides every exact Excel expression. Input units and modelling conventions appear in each workbook's Guide and Inputs sheets.

Statements B7 Opening Revenue

Mathematical equivalence: Opening value from Inputs B7

Units and scope: KES million

= 'Inputs' !B7

Statements B17 Opening Receivables

Mathematical equivalence: Opening value from Inputs B8

Units and scope: KES million

= 'Inputs' !B8

Statements B18 Opening Inventory

Mathematical equivalence: Opening value from Inputs B9

Units and scope: KES million

= 'Inputs' !B9

Statements B19 Opening Trade payables

Mathematical equivalence: Opening value from Inputs B10

Units and scope: KES million

= 'Inputs' !B10

Statements B25 Opening Closing plant

Mathematical equivalence: Opening value from Inputs B11

Units and scope: KES million

= 'Inputs' !B11

Statements B39 Opening Closing cash

Mathematical equivalence: Opening value from Inputs B12

Units and scope: KES million

= 'Inputs' !B12

Statements B29 Opening Closing debt

Mathematical equivalence: Opening value from Inputs B13

Units and scope: KES million

= 'Inputs' !B13

Statements B33 Opening Closing equity

Mathematical equivalence: Opening value from Inputs B14

Units and scope: KES million

= 'Inputs' !B14

Statements B20 Opening working capital

Mathematical equivalence: $NWC = \text{receivables} + \text{inventory} - \text{payables}$

Units and scope: KES million

= B17 + B18 - B19

Statements C7 Revenue

Mathematical equivalence: $\text{Revenue}_t = \text{Revenue}_{(t-1)} (1 + g_t)$

Units and scope: KES million; EPS and DPS KES/share

= B7 * (1 + 'Annual drivers' !B7)

Statements C8 Cash cost of sales

Mathematical equivalence: $COGS = \text{Revenue} \times \text{cost ratio}$

Units and scope: KES million; EPS and DPS KES/share

= C7 * 'Annual drivers' !B15

Statements C9 EBITDA

Mathematical equivalence: $EBITDA = \text{Revenue} \times \text{margin}$

Units and scope: KES million; EPS and DPS KES/share

=C7*'Annual drivers'!B8

Statements C10 Depreciation

Mathematical equivalence: $D\&A = \text{opening plant} \times \text{depreciation rate}$

Units and scope: KES million; EPS and DPS KES/share

=C23*'Annual drivers'!B9

Statements C11 EBIT

Mathematical equivalence: $EBIT = EBITDA - D\&A$

Units and scope: KES million; EPS and DPS KES/share

=C9-C10

Statements C12 Interest expense

Mathematical equivalence: $\text{Interest} = \text{opening debt} \times \text{rate}$

Units and scope: KES million; EPS and DPS KES/share

=C27*'Annual drivers'!B16

Statements C13 Profit before tax

Mathematical equivalence: $PBT = EBIT - \text{interest}$

Units and scope: KES million; EPS and DPS KES/share

=C11-C12

Statements C14 Cash tax expense

Mathematical equivalence: $\text{Tax} = \max(PBT, 0) \times \text{tax rate}$; no loss carryforwards

Units and scope: KES million; EPS and DPS KES/share

=MAX(0,C13)*'Annual drivers'!B11

Statements C15 Net profit

Mathematical equivalence: $NI = PBT - \text{tax}$

Units and scope: KES million; EPS and DPS KES/share

=C13-C14

Statements C17 Receivables

Mathematical equivalence: $AR = \text{Revenue} \times DSO / 365$

Units and scope: KES million; EPS and DPS KES/share

=C7*'Annual drivers'!B12/365

Statements C18 Inventory

Mathematical equivalence: $\text{Inventory} = \text{COGS} \times \text{DIO} / 365$

Units and scope: KES million; EPS and DPS KES/share

$= \text{C8} * \text{'Annual drivers'}!B13 / 365$

Statements C19 Trade payables

Mathematical equivalence: $\text{AP} = \text{COGS} \times \text{DPO} / 365$; COGS proxy for purchases

Units and scope: KES million; EPS and DPS KES/share

$= \text{C8} * \text{'Annual drivers'}!B14 / 365$

Statements C20 Operating working capital

Mathematical equivalence: $\text{NWC} = \text{AR} + \text{inventory} - \text{AP}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{C17} + \text{C18} - \text{C19}$

Statements C21 Change in working capital

Mathematical equivalence: $\Delta \text{NWC} = \text{NWC}_t - \text{NWC}_{(t-1)}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{C20} - \text{B20}$

Statements C23 Opening plant

Mathematical equivalence: $\text{Opening plant} = \text{prior closing plant}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{B25}$

Statements C24 Capital expenditure

Mathematical equivalence: $\text{Capex} = \text{Revenue} \times \text{capex fraction}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{C7} * \text{'Annual drivers'}!B10$

Statements C25 Closing plant

Mathematical equivalence: $\text{Plant}_{\text{close}} = \text{Plant}_{\text{open}} + \text{Capex} - \text{D\&A}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{C23} + \text{C24} - \text{C10}$

Statements C27 Opening debt

Mathematical equivalence: $\text{Debt}_{\text{open}} = \text{prior Debt}_{\text{close}}$

Units and scope: KES million; EPS and DPS KES/share

$= \text{B29}$

Statements C28 Principal repayment

Mathematical equivalence: $\text{Repayment} = \min(\text{opening debt}, \text{scheduled principal})$

Units and scope: KES million; EPS and DPS KES/share

$=\text{MIN}(\text{C27}, \text{'Annual drivers'!B17})$

Statements C29 Closing debt

Mathematical equivalence: $\text{Debt_close} = \text{Debt_open} - \text{repayment}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{C27} - \text{C28}$

Statements C31 Opening equity

Mathematical equivalence: $\text{Equity_open} = \text{prior Equity_close}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{B33}$

Statements C32 Dividends

Mathematical equivalence: $\text{Dividends} = \max(\text{NI}, 0) \times \text{payout}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{MAX}(0, \text{C15}) * \text{'Annual drivers'!B18}$

Statements C33 Closing equity

Mathematical equivalence: $\text{Equity_close} = \text{Equity_open} + \text{NI} - \text{dividends}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{C31} + \text{C15} - \text{C32}$

Statements C35 Opening cash

Mathematical equivalence: $\text{Cash_open} = \text{prior Cash_close}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{B39}$

Statements C36 Operating cash flow

Mathematical equivalence: $\text{CFO} = \text{NI} + \text{D\&A} - \Delta\text{NWC}$

Units and scope: KES million; EPS and DPS KES/share

$=\text{C15} + \text{C10} - \text{C21}$

Statements C37 Investing cash flow

Mathematical equivalence: $\text{CFI} = -\text{Capex}$

Units and scope: KES million; EPS and DPS KES/share

$= -\text{C24}$

Statements C38 Financing cash flow

Mathematical equivalence: $CFF = -\text{repayment} - \text{dividends}$

Units and scope: KES million; EPS and DPS KES/share

= -C28 - C32

Statements C39 Closing cash

Mathematical equivalence: $\text{Cash_close} = \text{Cash_open} + CFO + CFI + CFF$

Units and scope: KES million; EPS and DPS KES/share

= C35 + C36 + C37 + C38

Statements C41 Total assets

Mathematical equivalence: $\text{Assets} = AR + \text{Inventory} + \text{Plant} + \text{Cash}$

Units and scope: KES million; EPS and DPS KES/share

= C17 + C18 + C25 + C39

Statements C42 Liabilities plus equity

Mathematical equivalence: $\text{Liabilities} + \text{Equity} = AP + \text{Debt} + \text{Equity}$

Units and scope: KES million; EPS and DPS KES/share

= C19 + C29 + C33

Statements C43 Balance sheet difference

Mathematical equivalence: $\text{Check} = \text{Assets} - \text{Liabilities} - \text{Equity}$

Units and scope: KES million; EPS and DPS KES/share

= C41 - C42

Statements C45 NOPAT

Mathematical equivalence: $\text{NOPAT} = \text{EBIT} (1 - \text{tax rate})$

Units and scope: KES million; EPS and DPS KES/share

= C11 * (1 - 'Annual drivers'!B11)

Statements C46 Unlevered free cash flow

Mathematical equivalence: $\text{FCFF} = \text{NOPAT} + D\&A - \text{Capex} - \Delta \text{NWC}$

Units and scope: KES million; EPS and DPS KES/share

= C45 + C10 - C24 - C21

Statements C47 EPS

Mathematical equivalence: $\text{EPS} = \text{NI} / \text{shares}$

Units and scope: KES million; EPS and DPS KES/share

= C15 / 'Inputs'!\$B\$15

Statements C48 DPS

Mathematical equivalence: $DPS = \text{dividends} / \text{shares}$

Units and scope: KES million; EPS and DPS KES/share

=C32/'Inputs'!\$B\$15

DCF B7 PV forecast cash flow

Mathematical equivalence: $PV_FCFF = \sum FCFF_t / (1+WACC)^t$

Units and scope: KES million except per-share and fractions

=SUM('Statements'!C46/(1+'Inputs'!B16), 'Statements'!D46/(1+'Inputs'!B16)^2, 'Statements'!E46/(1+'Inputs'!B16)^3, 'Statements'!F46/(1+'Inputs'!B16)^4, 'Statements'!G46/(1+'Inputs'!B16)^5)

DCF B8 Terminal NOPAT

Mathematical equivalence: $NOPAT_6 = NOPAT_5(1+g)$

Units and scope: KES million except per-share and fractions

='Statements'!G45*(1+'Inputs'!B17)

DCF B9 Terminal reinvestment fraction

Mathematical equivalence: $RR = g / ROIC$

Units and scope: KES million except per-share and fractions

='Inputs'!B17/'Inputs'!B18

DCF B10 Terminal cash flow

Mathematical equivalence: $FCFF_6 = NOPAT_6(1-g/ROIC)$

Units and scope: KES million except per-share and fractions

=B8*(1-B9)

DCF B11 Terminal enterprise value

Mathematical equivalence: $TV_5 = FCFF_6 / (WACC - g)$

Units and scope: KES million except per-share and fractions

=IF('Inputs'!B16<='Inputs'!B17, "Invalid WACC <= growth", B10/('Inputs'!B16-'Inputs'!B17))

DCF B12 PV terminal

Mathematical equivalence: $PV_TV = TV_5 / (1+WACC)^5$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B11), B11/(1+'Inputs'!B16)^5, "Check terminal inputs")

DCF B13 Enterprise value

Mathematical equivalence: $EV = PV_FCFF + PV_TV$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B12), B7+B12, "Check terminal inputs")

DCF B14 Equity value

Mathematical equivalence: $\text{Equity} = \text{EV} + \text{cash} - \text{debt} + \text{investments} - \text{minority interests}$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B13),B13+'Inputs'!B12-'Inputs'!B13+'Inputs'!B20-'Inputs'!B19,"Check terminal inputs")

DCF B15 Value per share

Mathematical equivalence: $V = \text{Equity} / \text{shares}$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B14),B14/'Inputs'!B15,"Check terminal inputs")

DCF B16 Margin of safety

Mathematical equivalence: $\text{MOS} = 1 - \text{price} / \text{value}$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B15),1-'Inputs'!B21/B15,"Check terminal inputs")

DCF B17 Terminal share of EV

Mathematical equivalence: $\text{Terminal proportion} = \text{PV_TV} / \text{EV}$

Units and scope: KES million except per-share and fractions

=IF(ISNUMBER(B13),B12/B13,"Check terminal inputs")

DCF sensitivity B7 Value per share sensitivity

Mathematical equivalence:

$V(w, g) = [\sum \text{FCFF}_t / (1+w)^t + \text{NOPAT}_5(1+g)(1-g/\text{ROIC}) / (w-g) / (1+w)^5 + \text{cash} - \text{debt} + \text{investments} - \text{NCI}] / \text{shares}$

Units and scope: KES/share

=IF(\$A7<=B\$6,"Invalid",((('Statements'!C46/(1+\$A7)^1+'Statements'!D46/(1+\$A7)^2+'Statements'!E46/(1+\$A7)^3+'Statements'!F46/(1+\$A7)^4+'Statements'!G46/(1+\$A7)^5)+'Statements'!G45*(1+B\$6)*(1-B\$6/'Inputs'!B\$18)/(\$A7-B\$6)/(1+\$A7)^5+'Inputs'!B\$12-'Inputs'!B\$13+'Inputs'!B\$20-'Inputs'!B\$19)/'Inputs'!B\$15)

Bank scenarios L7 After-tax scenario earnings

Mathematical equivalence: $\text{NI} = (\text{pre-provision profit} - \text{credit loss})(1 - \text{tax})$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

=(I7-J7)*(1-K7)

Bank scenarios M7 Required retained capital

Mathematical equivalence: $\text{retained} = \max[0, \text{RWA}_0(1 + \text{growth}) \times \text{capital ratio} - \text{opening capital}]$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

=MAX(0,F7*(1+G7)*H7-B7)

Bank scenarios N7 Indicative dividend capacity

Mathematical equivalence: $\text{dividend} = \max(0, \text{NI} - \text{required retention})$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

$=\text{MAX}(0, \text{L7} - \text{M7})$

Bank scenarios O7 Scenario payout

Mathematical equivalence: $\text{payout} = \text{dividend} / \text{NI}$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

$=\text{IF}(\text{L7} > 0, \text{N7} / \text{L7}, \text{"Not meaningful"})$

Bank scenarios P7 Steady-state justified P/B

Mathematical equivalence: $\text{P/B} = (\text{ROE} - g) / (k_e - g)$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

$=\text{IF}(\text{OR}(\text{D7} \leq \text{E7}, \text{C7} < \text{E7}), \text{"Invalid growth"}, (\text{C7} - \text{E7}) / (\text{D7} - \text{E7}))$

Bank scenarios Q7 Steady-state equity value

Mathematical equivalence: $V = \text{B0}(\text{ROE} - g) / (k_e - g)$

Units and scope: Stable ROE valuation separate from one-year earnings stress.

$=\text{IF}(\text{ISNUMBER}(\text{P7}), \text{P7} * \text{B7}, \text{"Check growth"})$

Insurance scenarios K7 Loss ratio

Mathematical equivalence: $\text{LR} = \text{claims} / \text{earned premium}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=\text{C7} / \text{B7}$

Insurance scenarios L7 Expense ratio

Mathematical equivalence: $\text{ER} = \text{expenses} / \text{earned premium}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=\text{D7} / \text{B7}$

Insurance scenarios M7 Combined ratio

Mathematical equivalence: $\text{CR} = \text{LR} + \text{ER}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=\text{K7} + \text{L7}$

Insurance scenarios N7 Underwriting result

Mathematical equivalence: $U = \text{earned premium} - \text{claims} - \text{expenses}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=\text{B7} - \text{C7} - \text{D7}$

Insurance scenarios O7 Profit before tax

Mathematical equivalence: $PBT = U + \text{investment} + \text{other} - \text{finance}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=N7+E7+F7-G7$

Insurance scenarios P7 Profit after tax

Mathematical equivalence: $PAT = PBT - \max(PBT, 0) \times \text{tax}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=O7 - \text{MAX}(0, O7) * H7$

Insurance scenarios Q7 Average equity

Mathematical equivalence: $E_{\text{avg}} = (E_{\text{open}} + E_{\text{close}}) / 2$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=(I7+J7)/2$

Insurance scenarios R7 Return on equity

Mathematical equivalence: $ROE = PAT / E_{\text{avg}}$

Units and scope: No tax credit on losses; other comprehensive income may change closing equity.

$=P7/Q7$

Sector screens J7 Inventory days

Mathematical equivalence: $DIO = \text{average inventory} / \text{COGS} \times 365$

Units and scope: KES million except days and cover multiple

$=\text{AVERAGE}(D7:E7) / C7 * 365$

Sector screens K7 Collection days

Mathematical equivalence: $DSO = \text{average AR} / \text{revenue} \times 365$

Units and scope: KES million except days and cover multiple

$=\text{AVERAGE}(F7:G7) / B7 * 365$

Sector screens L7 Payable days

Mathematical equivalence: $DPO = \text{average AP} / \text{COGS} \times 365$

Units and scope: KES million except days and cover multiple

$=\text{AVERAGE}(H7:I7) / C7 * 365$

Sector screens M7 Cash conversion cycle

Mathematical equivalence: $CCC = DIO + DSO - DPO$

Units and scope: KES million except days and cover multiple

$=J7+K7-L7$

Sector screens P7 Cash after capex

Mathematical equivalence: $FCF_proxy = CFO - capex$

Units and scope: KES million except days and cover multiple

=N7-O7

Sector screens R7 Cash dividend cover

Mathematical equivalence: $cover = FCF_proxy / dividend$

Units and scope: KES million except days and cover multiple

=IF(Q7>0,P7/Q7,"No dividend")

Bank diagnostics P7 NPL ratio

Mathematical equivalence: $NPL / gross\ loans$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=C7/B7

Bank diagnostics Q7 Allowance coverage

Mathematical equivalence: $allowance / NPL$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=D7/C7

Bank diagnostics R7 Cost of risk

Mathematical equivalence: $annual\ credit\ loss / average\ gross\ loans$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=E7/AVERAGE(B7,F7)

Bank diagnostics S7 Net interest margin

Mathematical equivalence: $NII / average\ earning\ assets$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=G7/H7

Bank diagnostics T7 Cost-to-income

Mathematical equivalence: $operating\ expense / operating\ income$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=I7/J7

Bank diagnostics U7 Core capital ratio

Mathematical equivalence: $core\ capital / RWA$

Units and scope: Fractions; headroom in percentage points when multiplied by100.

=K7/M7

Bank diagnostics V7 Total capital ratio

Mathematical equivalence: $\text{total capital}/\text{RWA}$

Units and scope: Fractions; headroom in percentage points when multiplied by 100.

$$=L7/M7$$

Bank diagnostics W7 Core headroom

Mathematical equivalence: $\text{core ratio}-\text{required ratio}$

Units and scope: Fractions; headroom in percentage points when multiplied by 100.

$$=U7-N7$$

Bank diagnostics X7 Total headroom

Mathematical equivalence: $\text{total ratio}-\text{required ratio}$

Units and scope: Fractions; headroom in percentage points when multiplied by 100.

$$=V7-O7$$

Historical quality K7 Return on assets

Mathematical equivalence: $\text{NI}/\text{closing assets}$, same basis each year

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

$$=D7/F7$$

Historical quality L7 Cash return on assets

Mathematical equivalence: $\text{CF0}/\text{closing assets}$

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

$$=E7/F7$$

Historical quality M7 Gross margin

Mathematical equivalence: $(\text{Revenue}-\text{COGS})/\text{Revenue}$

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

$$=(B7-C7)/B7$$

Historical quality N7 Asset turnover

Mathematical equivalence: $\text{Revenue}/\text{closing assets}$

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

$$=B7/F7$$

Historical quality O7 Leverage

Mathematical equivalence: $\text{Long-term debt}/\text{closing assets}$

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

$$=G7/F7$$

Historical quality P7 Current ratio

Mathematical equivalence: $\text{Current assets} / \text{current liabilities}$

Units and scope: Comparable classroom ratios; full Piotroski research conventions use defined asset timing.

=H7/I7

Quality signals D7 Positive ROA

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B7>C7,1,0)

Quality signals D8 Positive cash flow

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B8>C8,1,0)

Quality signals D9 Improving ROA

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B9>C9,1,0)

Quality signals D10 Cash exceeds accrual profit

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B10>C10,1,0)

Quality signals D11 Lower leverage

Mathematical equivalence: $\text{point} = 1\{\text{current} < \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B11<C11,1,0)

Quality signals D12 Improving liquidity

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B12>C12,1,0)

Quality signals D13 No share issuance

Mathematical equivalence: $\text{point} = 1\{\text{current} \leq \text{comparison}\}; \text{else } 0$

Units and scope: Binary signal; see stated denominator conventions.

=IF(B13<=C13,1,0)

Quality signals D14 Improving gross margin

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{ else } 0$

Units and scope: Binary signal; see stated denominator conventions.

$=\text{IF}(\text{B14}>\text{C14},1,0)$

Quality signals D15 Improving asset turnover

Mathematical equivalence: $\text{point} = 1\{\text{current} > \text{comparison}\}; \text{ else } 0$

Units and scope: Binary signal; see stated denominator conventions.

$=\text{IF}(\text{B15}>\text{C15},1,0)$

Quality signals D17 Combined educational signal count

Mathematical equivalence: $\text{score} = \sum 9 \text{ binary points}$

Units and scope: 0–9; research prioritisation, not automatic selection

$=\text{SUM}(\text{D7:D15})$